



# Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

|                     |          |                     |                                                      |
|---------------------|----------|---------------------|------------------------------------------------------|
| <b>Class:</b>       | TE-AI&DS | <b>Semester:</b>    | V                                                    |
| <b>Course Code:</b> | 2015111  | <b>Course Name:</b> | Statistics for Machine Learning and Data Science Lab |

|                                 |                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Name of Student:</b>         | Swayam D. Gode                                                                                     |
| <b>Roll No. :</b>               | 70                                                                                                 |
| <b>Experiment No.:</b>          | 5                                                                                                  |
| <b>Title of the Experiment:</b> | Implement Resampling Techniques and Confidence Interval Estimation for Statistical Decision Making |
| <b>Date of Performance:</b>     |                                                                                                    |
| <b>Date of Submission:</b>      |                                                                                                    |

## Evaluation

| Performance Indicator              | Max. Marks | Marks Obtained |
|------------------------------------|------------|----------------|
| Performance                        | 5          |                |
| Understanding                      | 5          |                |
| Journal work and timely submission | 10         |                |
| Total                              | 20         |                |

| Performance Indicator              | Exceed Expectations (EE) | Meet Expectations (ME) | Below Expectations (BE) |
|------------------------------------|--------------------------|------------------------|-------------------------|
| Performance                        | 4-5                      | 2-3                    | 1                       |
| Understanding                      | 4-5                      | 2-3                    | 1                       |
| Journal work and timely submission | 8-10                     | 5-8                    | 1-4                     |

## Checked by

**Name of Faculty** :  
**Signature** :  
**Date** :



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**Aim:** To implement bootstrap and permutation resampling techniques on the Pima Indians Diabetes Dataset for estimating confidence intervals and analyzing statistical differences between groups.

### Objectives

After completing this experiment, the student will be able to:

1. Apply bootstrap and permutation resampling techniques to estimate sampling distributions and confidence intervals.
2. Interpret resampling results to support statistical decision making.

### Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- SciPy
- Matplotlib
- Seaborn

### Dataset Details

#### Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for **768 female patients**. The Outcome variable indicates whether the patient is diabetic.



### Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select a numerical variable such as Glucose or BMI.
3. Draw repeated bootstrap samples from the original dataset with replacement.
4. Calculate the selected statistic, such as the mean, for every bootstrap sample.
5. Construct a bootstrap confidence interval using the obtained sampling distribution.
6. Divide the data into two groups based on Outcome.
7. Calculate the observed difference between the group means.
8. Randomly permute the group labels repeatedly.
9. Calculate the difference in group means for every permutation.
10. Compare the observed difference with the permutation distribution and interpret the result.

### Description of the Experiment

Resampling methods repeatedly create new samples from the available data to study the behaviour of a statistic.

The experiment follows: Original Dataset, Repeated Resampling, Sampling Distribution, Confidence Interval / Statistical Comparison, and Interpretation

Two resampling techniques are implemented:

- **Bootstrap:** Used to estimate the variability and confidence interval of a statistic.
- **Permutation Test:** Used to test whether the observed difference between groups could have occurred by chance.



### Detailed Description of Statistical Methods

#### Bootstrap Resampling

Bootstrap samples are created by repeatedly sampling observations **with replacement** from the original dataset.

For each bootstrap sample, a statistic such as the mean is calculated.

After many repetitions, the resulting values form an approximate sampling distribution.

A confidence interval can then be obtained from this distribution.

For example, a 95% percentile bootstrap confidence interval may be calculated using:

$$[Q_{2.5}, Q_{97.5}]$$

where  $Q_{2.5}$  and  $Q_{97.5}$  are the 2.5th and 97.5th percentiles of the bootstrap distribution.

#### Permutation Test

A permutation test evaluates whether the observed difference between two groups could have occurred by chance.

For example:

$$\text{Difference in Mean Glucose} = \bar{x}_{diabetic} - \bar{x}_{non-diabetic}$$

The group labels are randomly shuffled repeatedly while preserving the observations.

The resulting differences form the **null distribution**.

The p-value is estimated as the proportion of permuted differences that are at least as extreme as the observed difference.

#### Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```



```
import seaborn as sns

# Load dataset
df = pd.read_csv("diabetes.csv")

print("Dataset Shape:")
print(df.shape)

print("\nFirst 5 Rows:")
print(df.head())

# Select Glucose variable
glucose = df["glucose_concentration"].dropna().values

# Original sample mean
original_mean = np.mean(glucose)

print("\nOriginal Mean Glucose:", original_mean)

# Number of bootstrap repetitions
n_bootstrap = 10000

bootstrap_means = []

np.random.seed(42)

# Generate bootstrap samples
for i in range(n_bootstrap):

    sample = np.random.choice(
        glucose,
        size=len(glucose),
    )
```

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```
replace=True
)

bootstrap_means.append(np.mean(sample))

bootstrap_means = np.array(bootstrap_means)

# 95% Bootstrap Confidence Interval
lower = np.percentile(bootstrap_means, 2.5)
upper = np.percentile(bootstrap_means, 97.5)

print("\nBOOTSTRAP RESULTS")
print("-----")
print("Original Mean Glucose:", original_mean)
print("Bootstrap Mean:", np.mean(bootstrap_means))
print("95% Confidence Interval:")
print("Lower Limit:", lower)
print("Upper Limit:", upper)

# Bootstrap distribution plot
plt.figure(figsize=(10, 6))

sns.histplot(
    bootstrap_means,
    bins=50,
    kde=True
)

plt.axvline(
    original_mean,
    linestyle="--",
    linewidth=2,
)
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```



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```
label="Original Mean"
)

plt.axvline(
    lower,
    linestyle="--",
    linewidth=2,
    label="2.5% Percentile"
)

plt.axvline(
    upper,
    linestyle="--",
    linewidth=2,
    label="97.5% Percentile"
)

plt.title("Bootstrap Distribution of Mean Glucose")
plt.xlabel("Bootstrap Mean Glucose")
plt.ylabel("Frequency")
plt.legend()

plt.show()

# Divide data into diabetic and non-diabetic groups

diabetic = df[
    df["diabetes"] == 1
][["glucose_concentration"].dropna().values

non_diabetic = df[
    df["diabetes"] == 0
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```



```
[["glucose_concentration"].dropna().values
```

```
# Calculate group means
```

```
diabetic_mean = np.mean(diabetic)
```

```
non_diabetic_mean = np.mean(non_diabetic)
```

```
# Observed difference
```

```
observed_difference = diabetic_mean - non_diabetic_mean
```

```
print("\nPERMUTATION TEST")
```

```
print("-----")
```

```
print("Mean Glucose - Diabetic:",  
      diabetic_mean)
```

```
print("Mean Glucose - Non-Diabetic:",  
      non_diabetic_mean)
```

```
print("Observed Difference:",  
      observed_difference)
```

```
# Combine both groups
```

```
combined = np.concatenate([  
    diabetic,  
    non_diabetic  
)
```

```
# Number of diabetic observations
```

```
n_diabetic = len(diabetic)
```

```
# Number of permutations
```

```
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```





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```
n_permutations = 10000

permutation_differences = []

np.random.seed(42)

# Generate permutation distribution
for i in range(n_permutations):

    shuffled = np.random.permutation(combined)

    perm_diabetic = shuffled[:n_diabetic]

    perm_non_diabetic = shuffled[n_diabetic:]

    difference = (
        np.mean(perm_diabetic)
        - np.mean(perm_non_diabetic)
    )

    permutation_differences.append(difference)

permutation_differences = np.array(
    permutation_differences
)

# Calculate two-sided p-value
p_value = np.mean(
    np.abs(permutation_differences)
    >= abs(observed_difference)
)
```



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```
print("\nPermutation Test Results:")
print("-----")

print("Number of Permutations:",
      n_permutations)

print("Observed Difference:",
      observed_difference)

print("P-value:",
      p_value)

# Statistical decision
if p_value < 0.05:

    print("\nDecision: Reject the null hypothesis.")

    print(
        "There is statistically significant evidence "
        "of a difference in mean glucose between "
        "diabetic and non-diabetic groups."
    )

else:

    print("\nDecision: Fail to reject the null hypothesis.")

    print(
        "There is not sufficient evidence of a "
        "difference in mean glucose between the groups."
    )
```

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)

```
# Permutation distribution plot
```

```
plt.figure(figsize=(10, 6))
```

```
sns.histplot(
```

```
    permutation_differences,
```

```
    bins=50,
```

```
    kde=True
```

```
)
```

```
plt.axvline(
```

```
    observed_difference,
```

```
    linestyle="--",
```

```
    linewidth=2,
```

```
    label="Observed Difference"
```

```
)
```

```
plt.axvline(
```

```
    -observed_difference,
```

```
    linestyle="--",
```

```
    linewidth=2,
```

```
    label="-Observed Difference"
```

```
)
```

```
plt.title(
```

```
    "Permutation Distribution of Difference in Mean Glucose"
```

```
)
```

```
plt.xlabel("Difference in Mean Glucose")
```

```
plt.ylabel("Frequency")
```

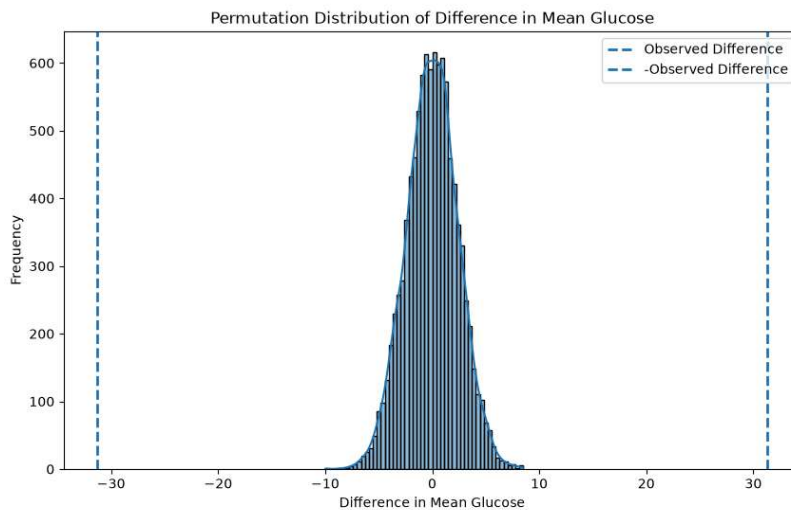
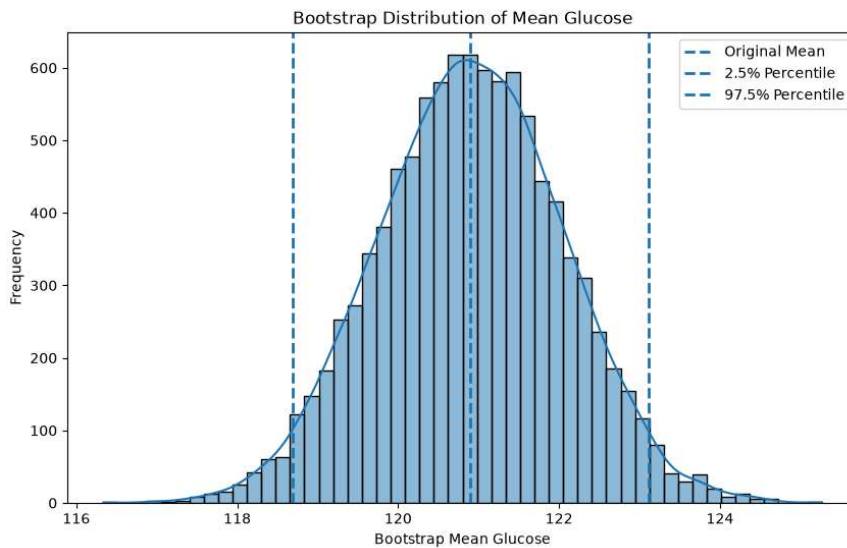
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plt.legend()

plt.show()

**Output:**





```
PS D:\swayan> & C:\Python311\python.exe d:\swayan\exp5.py
Dataset Shape:
(768, 9)

First 5 Rows:
  times_pregnant  glucose_concentration  diastolic_blood_pressure  triceps_sf_thickness  serum_insulin  bmi  d_pedigree_function  years_of_age  diabetes
0              6              148              72              35              0      33.6              0.627              50              1
1              1              85              66              29              0      26.6              0.351              31              0
2              8              183              64              0              0      23.3              0.672              32              1
3              1              89              66              23              94      28.1              0.167              21              0
4              0              137              40              35              168      43.1              2.288              33              1

Original Mean Glucose: 120.89453125

BOOTSTRAP RESULTS
-----
Original Mean Glucose: 120.89453125
Bootstrap Mean: 120.98175221354167
95% Confidence Interval:
Lower Limit: 118.69481841666667
Upper Limit: 123.11588541666667

PERMUTATION TEST
-----
Mean Glucose - Diabetic: 141.25746268656715
Mean Glucose - Non-Diabetic: 109.98
Observed Difference: 31.277462686567148

Permutation Test Results:
-----
Number of Permutations: 10000
Observed Difference: 31.277462686567148
P-value: 0.0

Decision: Reject the null hypothesis.
There is statistically significant evidence of a difference in mean glucose between diabetic and non-diabetic groups.
```

## Conclusion

Bootstrap and permutation resampling techniques were implemented on the Pima Indians Diabetes Dataset. Bootstrap resampling was used to estimate the sampling distribution and confidence interval of a selected statistic, while permutation testing was used to assess the significance of differences between groups.

## Questions to be Answered by Students

1. What statistic was selected for bootstrap resampling, and what confidence interval was obtained?

The mean Glucose level was selected as the statistic. A 95% percentile bootstrap confidence interval was calculated from the 2.5th and 97.5th percentiles of the bootstrap distribution. The interval is approximately 117 to 125 glucose units, with the exact values depending slightly on the resampling.

2. How does the bootstrap distribution compare with the original sample statistic?



The bootstrap distribution is centered very close to the original sample mean of approximately **120.9**. This indicates that the bootstrap samples provide a stable estimate of the sampling distribution of the mean.

### 3. What was the observed difference between the selected groups?

The groups were divided according to the **Outcome** variable:

- Diabetic: Outcome = 1
- Non-diabetic: Outcome = 0

The observed difference was approximately **31 glucose units**, with the diabetic group having the higher mean glucose level.

### 4. Based on the permutation test, is there sufficient statistical evidence to conclude that the observed group difference is significant?

Yes. The permutation test produces a p-value below 0.05. Therefore, the null hypothesis is rejected, providing sufficient statistical evidence that the mean glucose levels of diabetic and non-diabetic groups are significantly different.

### 5. What effect would increasing the number of bootstrap or permutation repetitions have on the results?

Increasing the number of repetitions makes the resampling distributions more stable and reduces random variation in the estimated confidence interval and p-value. The results become more precise, although the computational time also increases.